

18 **Ans: A**

$$L_x = 2L_y$$

$$A_x = \pi r_x^2 = \pi(0.5r_y)^2 = 0.25\pi r_y^2 = 0.25 A_y$$

Since resistance are equal,

$$\frac{\frac{\rho_x L_x}{A_x}}{\frac{\rho_y L_y}{A_y}} = 1$$

$$\frac{\frac{\rho_x (2L_y)}{0.25 A_y}}{\frac{\rho_y L_y}{A_y}} = 1$$

$$\frac{8\rho_x}{\rho_y} = 1$$

$$\frac{\rho_x}{\rho_y} = \frac{1}{8}$$

